

Basepoint-Relative Stability and Codomain Drift: A Heap-Theoretic Account

****Formalization of claims C001?C010 (Lean 4 / mathlib)****

2026-08-12 · Internal research paper · Lean 4 / mathlib v4.32.2

Abstract

We develop a heap-theoretic account of how the choice of a basepoint stabilizes relative structure.

1. Introduction

1.1 Motivation

In mathematics, a structure often becomes a **pointed** object only after the choice of a basepoint.

1.2 Contributions

1. ****Heap retract baseline (C004)****: an abelian heap recovers an abelian group at any basepoint
2. ****Transport under basepoint change (C006)****: $T_{\{e?f\}}(x) = [x, e, f]$ is an automorphism on an abelian heap
3. ****Displacement space (C007)****: the equivalence $(e,a) \sim (f,b) \iff b = [a, e, f]$ is an equivalence on $H \times H$
4. ****Endogenous generation law (C008?C010)****: a generation law $\gamma : e \rightarrow ?_e$ on pure-heap terms y is endogenous iff $\gamma y = [y, e, ?_e]$
5. ****Nature of natural-number-like structure (C009)****: finitary reachability (TransGen) presupposes a generation law

1.3 Honest boundary

All results are known in classical heap/torsor theory (novelty: KNOWN for C004; DIRECT_COROLLARY for C006).

2. Preliminaries: heaps and basepoint retraction

****Definition 2.1**** (heap). A heap is a set H with a ternary operation $[_, _, _] : H \times H \times H \rightarrow H$ satisfying:

****Theorem 2.2**** (heap retract, C004, KNOWN). Let H be an abelian heap and $e : H$ a chosen basepoint. Then $[x, e, _]$ is a retraction of H to $\{e\}$.

****Observation 2.3**** (codomain drift). The choice of basepoint is not unique: for $e \sim f$, the retraction $[x, e, _]$ is conjugate to $[x, f, _]$ via $[e, e, _]$.

3. Transport under basepoint change (C006)

****Definition 3.1**** (basepoint-change map). For a heap H and basepoints $e, f : H$, define

$$T_{\{e?f\}}(x) := [x, e, f].$$

****Theorem 3.2**** (T is an automorphism, C006). On any heap H (not necessarily abelian), $T_{\{e?f\}}$ is an automorphism of H .

$$T_{\{e?f\}}([x, y, z]) = [T_{\{e?f\}}(x), T_{\{e?f\}}(y), T_{\{e?f\}}(z)].$$

Proof: the middle para-associativity law of heaps makes the translation a homomorphism. Commutativity follows from the definition of T .

****Theorem 3.3**** (transport of step translation, C006). For fixed endpoint a , the step translation $[a, _, _]$ is transported to $[a, T_{\{e?f\}}(_), _]$ via $T_{\{e?f\}}$.

$$T_{\{e?f\}}([a, x, _]) = [a, T_{\{e?f\}}(x), _].$$

The step at e with endpoint a , viewed from f , is the step at f with the transported endpoint $T_{\{e?f\}}(a)$.

****Corollary 3.4**** (group isomorphism). The basepoint change induces an isomorphism between the groups $(H, [e, _, _])$ and $(H, [f, _, _])$.

4. Displacement space: relative stability (C007)

****Definition 4.1**** (displacement equivalence). On $H \times H$, define $(e, a) \sim (f, b)$ iff $b = [a, e, f]$.

****Theorem 4.2**** (equivalence, C007). $\sim$ is an equivalence relation on $H \times H$ (reflexive, symmetric, transitive).

****Definition 4.3**** (displacement space). Let $D(H) := (H \times H) / \sim$ be the quotient. An element of $D(H)$ is a displacement $[e, a]$.

****Theorem 4.4**** (recovery, C007). Choosing any basepoint e gives a bijection $[e, _] : D(H) \rightarrow G_e$.

****Corollary 4.5**** (the dichotomy). ****Basepoint choice stabilizes relative structure** ($D(H)$ is e -invariant) iff $(H, [e, _, _])$ is a group.

5. Endogenous generation (C008?C010)

5.1 Methodological correction: no presupposed counting (C008, C009)

****Definition 5.1**** (finitary reachability, C008, baseline). Relation $\text{TransGen } r \ a \ b$ means that a can be reached from b by a finite sequence of steps $[x, _, _]$ where $_$ is in the image of r .

